

AthDGC · Athens-PROIEL

An open diachronic Greek treebank with Indo-European parallels

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The **Athens Diachronic Glossa Chronos** (*Athens-PROIEL (the dependency-treebank (a collection of sentences whose grammatical structure has been analysed and stored) standard for early Indo-European languages, developed at Oslo)*) is an open PROIEL-style dependency treebank spanning Greek diachrony with Indo-European parallels.

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AthDGC is **both** a continuously-updated **PROIEL XML 2.0** (the file format that stores each sentence as a tree of word-by-word grammatical relations, developed at Oslo) treebank **and** an open-source **computational diachronic-linguistics toolkit**:

- LightSIDE (an open-source text-mining workbench developed at Carnegie Mellon)-compatible feature extraction + classifier training
- NoSketch-style concordancer (a search tool that shows every occurrence of a word in its surrounding context)
- Fine-tuned (further trained on new data to adapt it to a particular text type) Stanza (Stanford's open-source Python pipeline that automatically tags, lemmatises, and parses sentences) checkpoints (Byzantine, Late-Byzantine, Modern adaptations)
- Argument-structure extractor
- Cross-lingual alignment (matching corresponding words or sentences between

Covers:

- Homeric and Archaic Greek (Homer, Hesiod, Sappho, Pindar)
- Classical (Aeschylus, Sophocles, Euripides, Aristophanes, Plato, Aristotle)
- Koine and NT (Plutarch, Strabo, Lucian, the New Testament)
- Late Antique (Eusebius, Basil, Chrysostom, Procopius)
- Byzantine (Psellos, Anna Komnene, John of Damascus, Niketas Choniates)

with **NT verse-level cross-alignment** to Latin (Vulgate), Gothic (Wulfila), and Old Church Slavonic (Marianus).

Metric	Value
Total corpus rows	89.9 M
Annotated Greek rows	4.08 M
NT-aligned Greek verses	6,861
IE cross-aligned witnesses	4

Pipeline

1. **Discovery** - daily harvest from archive.org, Perseus, First1K, Wikisource, Diorisis
2. **Filtering** - Greek-script ratio, apparatus-criticus rejection, dedup
3. **Conversion** - PROIEL XML 2.0
4. **Annotation** - Stanza `grc_proiel` + analogous models per parallel language
5. **Argument-structure** - subject, object, oblique, voice, aspect per verb
6. **Cross-alignment** - LaBSE (a multilingual sentence-embedding model that maps sentences from many languages into a common vector space) + AwesomeAlign (a word-alignment procedure that uses multilingual-BERT attention to match words across translations) (mBERT)
7. **Storage** - PROIEL XML + JSONL (a plain-text data format that stores one record per line) + Qdrant + Neo4j

- **Retranslation** of influential texts (Iliad, NT, Septuagint Psalms, classical historiography)
- **Retelling** chains across periods and languages
- **Argument structure** (the set of obligatory and optional partners a verb requires, e.g. subject, object, oblique) under retranslation

- Lavidas (PI) · Nikiforidou · Haug (Oslo) · Kulikov (Ghent)
- Geka · Symeonidis · Michalareas
- Chionidi · Tsiropina · Plakoutsi · Argyropoulos

NKUA · Athens Digital Glossa Chronos Research Network

Lavidas, N., Nikiforidou, K., Haug, D., Kulikov, L., Geka, V., Symeonidis, V., Michalareas, T., Chionidi, S., Tsiropina, A., Plakoutsi, E., Argyropoulos, E., and the Athens Digital Glossa Chronos Research Network (2026). *AthDGC: Athens Diachronic Glossa Chronos*. Zenodo (an open research-data repository hosted at CERN that mints permanent DOIs).

DOI (Digital Object Identifier, a permanent web link for a published artefact):
10.5281/zenodo.20439182

- Showcase: <https://athdgc.github.io>
- Samples: <https://athdgc.github.io/samples.html>
- Tools: <https://athdgc.github.io/tools.html>
- Press kit: <https://athdgc.github.io/press.html>
- Launch report: https://athdgc.github.io/data_paper.html
- GitHub: <https://github.com/AthDGC>
- Hugging Face (a public hosting platform for machine-learning models and datasets) mirror: <https://huggingface.co/AthDGC> (in setup; ships at v0.5)

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